

Sukhrob Ilyosbekov

Computer Vision · Deep Learning · Explainable AI

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RESEARCH INTERESTS

I want vision models that people can trust enough to actually deploy. Three questions keep coming up in my work. Can a clinician check what the model looked at, instead of taking a score on faith? How much of a black-box generative model's behavior can you control from the outside, without retraining it? And do learned representations hold up on scientific imaging, where the data looks nothing like ImageNet and the labels are noisy? I also build ML for a HIPAA-regulated clinical platform, so I run into the same questions with real patients on the other end.

EDUCATION

Northeastern University, Boston, MA

January 2024 – May 2026

- Master of Science in Computer Science; GPA: 4.0/4.0
- Relevant Coursework: CS 7180 Advanced Perception, Machine Learning, Computer Vision

Suffolk University, Boston, MA

September 2021 – May 2023

- Bachelor of Science in Computer Science; Cum Laude; GPA: 3.5/4.0
- Transferred from INTI International College, Malaysia (2019 – 2021) and Tashkent University of Information Technologies, Uzbekistan (2017 – 2019)

PUBLICATIONS

- [1] **S. Ilyosbekov**, S. Gajjar, and R. Jin, “MorphoCLIP: Text-Supervised Contrastive Learning for Perturbation Matching in Cell Painting Images,” *arXiv preprint*, arXiv:2608.22690, 2026. [\[Paper\]](#)
- [2] **S. Ilyosbekov**, “Localize, Don't Beautify: Client-Side Control of Image-Editing APIs for Cosmetic Surgery Previews,” *arXiv preprint*, arXiv:2608.02841, 2026. [\[Paper\]](#)
- [3] **S. Ilyosbekov**, “MelanomaNet: Explainable Deep Learning for Multi-Class Skin Lesion Classification,” *arXiv preprint*, arXiv:2512.09289, 2025. [\[Paper\]](#)

AWARDS & HONORS

- **National Physics Olympiad, 1st Place**, Uzbekistan; first student from my high school to win nationally
- **Associate Member**, Sigma Xi, The Scientific Research Honor Society (2026)
- **University Coding Competition Winner**, Tashkent University of Information Technologies
- **Outstanding Project Award**, Bookshelf Scanner, Computer Vision Course, Northeastern University

RESEARCH

MorphoCLIP: Text-Supervised Cell Painting Retrieval

[GitHub](#) | [Paper](#)

- Built a text-supervised contrastive model that matches Cell Painting microscopy images of drug- and gene-perturbed cells to descriptions of the treatment written in ordinary language. The vision and language backbones (DINOv3, BioClinical ModernBERT) stay frozen and only small projection heads train, which is what keeps the whole thing on a single consumer GPU.
- Tested retrieval in both directions, image-to-text and text-to-image, on the CPJUMP1 benchmark: 51 plates, 3M+ cell images, 303 drugs, 160 genes. Batch correction in the embedding space keeps the model from simply learning which plate an image came from.
- Collaborative work with Northeastern co-authors S. Gajjar and R. Jin; I led the model design, training, and evaluation.
- **Stack**: PyTorch, DINOv3, BioClinical ModernBERT, contrastive learning, CPJUMP1

Localize, Don't Beautify: Client-Side Control of Image-Editing APIs

[Paper](#)

- Ask a commercial image editor to change one feature of a face and it beautifies the whole thing. Cosmetic-surgery previews are where that goes wrong most visibly, which makes them a good test case for how much of a black-box model's behavior you can control from the outside.
- Compared prompt-only steering, masked compositing, and model-based inpainting across six commercial editors and one inpainting model, over 196 facelift and rhinoplasty edits, scored on identity preservation (ArcFace) and on whether the edit stayed where it was asked to. Plain masked compositing came out ahead of model-based inpainting on localization.
- **Stack**: Image-editing APIs, inpainting models, ArcFace, prompt engineering

MelanomaNet: Explainable Deep Learning for Skin Lesion Classification

[GitHub](#) | [Paper](#)

- Trained an EfficientNet V2 classifier across all 9 ISIC 2019 diagnostic categories at 384×384 , with focal loss to cope with the heavy class imbalance. It reaches 85.6% accuracy and 0.856 weighted F1 on 25,331 dermoscopic images.
- Broke GradCAM++ attention down along the ABCDE criteria dermatologists already use, with asymmetry, border irregularity, color variation (K-means), and diameter measured straight off the lesion mask.
- Because those clinical features come out as numbers, how well they line up with the model's attention can be scored. The interpretability claim rests on a metric instead of a few good-looking heatmaps.
- **Stack:** PyTorch, EfficientNet V2, GradCAM++, OpenCV, focal loss, ISIC 2019

SELECTED COURSE PROJECTS

LightDepth: Lightweight Monocular Depth Estimation

[GitHub](#)

- Designed a ResNet18 encoder with a U-Net decoder and skip connections for monocular depth. It holds accuracy at 14.3M parameters where Depth Anything V2 needs 24.8M, a 42% cut.
- Inference runs 72% faster, and relative error on NYU Depth V2 improved too: 5.3% on AbsRel, 31.1% on SqRel.
- **Stack:** PyTorch, ResNet18, U-Net, NYU Depth V2

FSRCNN: Accelerating Super-Resolution Convolutional Neural Network

[GitHub](#)

- Reproduced FSRCNN (Dong et al., ECCV 2016) for single-image super-resolution at $2\times$, $3\times$, and $4\times$. Upsampling is learned end to end, which is where the $40\times$ speedup over SRCNN comes from.
- The reported gains held up on Set5 (+1.78 dB PSNR, +0.0507 SSIM) and Set14 (+1.26 dB PSNR, +0.0477 SSIM). Added ablations on the shrinking and mapping layers.
- **Stack:** PyTorch, mixed-precision training, T91/Set5/Set14/DIV2K

Bookshelf Scanner: Multi-Modal Book Detection and Recognition

[GitHub](#)

- Point a camera at a bookshelf and get back a list of what is on it. YOLO instance segmentation cuts out each spine, then the Moondream2 vision-language model reads the title and author off it. Won the course Outstanding Project Award.
- **Stack:** YOLO, Moondream2 VLM, llama.cpp, PyTorch, FastAPI

WORK EXPERIENCE

Senior AI/ML Engineer | [EmTech Care Labs](#) | Portland, ME

Jan 2025 – Present

- Built the machine-learning features for a HIPAA-compliant Alzheimer's care platform. The main one is a decline-risk model over daily activity and care-log data for roughly 1,200 residents; it holds 0.81 AUC and tends to flag someone about three weeks before the care team escalates on its own.
- Added an NLP pipeline that reads free-text clinical notes and pulls out structured details (symptoms, medications, care events) at roughly 0.90 F1 on a held-out set. Also stood up model serving and monitoring on de-identified data, under HIPAA constraints.

Machine Learning Engineer | [AllFactors](#) | Santa Clara, CA

Oct 2021 – Dec 2024

- Built property-valuation and market-forecasting models over roughly 2M listings across 40+ metros. Median absolute valuation error went from about 9% to 5% against the rules-based estimate the product had shipped with.
- Designed the real-time feature pipeline that fed live market data to the models and pushed updated predictions to users, then productionized the models into the customer-facing dashboards.

Earlier: Software Engineer at Smart Meal Service, Russia (2020 – 2021), control software for self-service food kiosks. Game Developer at Pentalight Technology, Malaysia (2020 – 2021), multiplayer networking and hand tracking for a VR smart-city simulation. Freelance Developer (2019 – 2020), Python NLP and .NET applications. Game Developer at EC Dev Team, Uzbekistan (2016 – 2019).

TECHNICAL SKILLS

Deep Learning & CV: PyTorch, TensorFlow/Keras, OpenCV, CNNs, Vision Transformers, U-Net, ResNet, EfficientNet, YOLO, contrastive and self-supervised learning, GradCAM++, classification, detection and segmentation, depth estimation, super-resolution, explainable AI

Mathematics: Linear algebra, probability and statistics, convex optimization, numerical methods, information theory, differential equations, hypothesis testing and experimental design

Machine Learning: Supervised and unsupervised learning, transfer learning, loss design, regularization, hyperparameter search, cross-validation and error analysis, XGBoost, K-means, NLP

Research Computing: Python, NumPy/SciPy, pandas, scikit-learn, Hugging Face, CUDA, mixed-precision and multi-GPU training, Weights & Biases, Docker, $\text{\LaTeX}$

Software Engineering: C#, C++, TypeScript; .NET, FastAPI, PostgreSQL, Kubernetes, AWS, Azure